This template is 36”x48” and can be printed on paper to hang up or mounted on foam core for display. An example on how to customize this template is on the next slide.

Printer – Menlo Park: American printing is a SLAC vendor that can invoice your department. Please check with your financial BP on the process for payment.

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COPY AND PASTE TO KEEP THE FORMATTING IN THIS TEMPLATE

On Mac:

Command+C (copy)

Ctl+Command+V (paste and select without formatting)

On PC:

Ctl+C (copy)

Ctl+Alt+V (paste and select without formatting)

LOGOS

To find a PNG logo required to be in white, most organizations have an identity use page with downloads. If you cannot find one, there is usually a contact email to request the file.

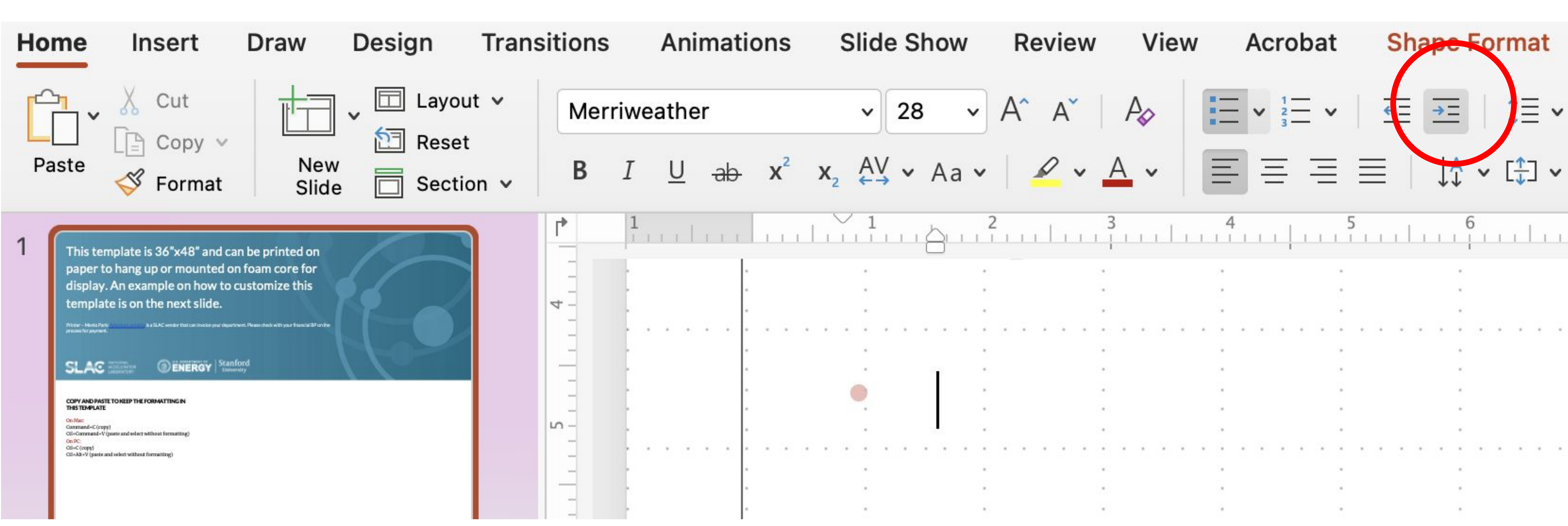
AN EXAMPLE AND YOUR TEMPLATE

On slide 2 you will find a sample of how to use this template to build your poster presentation.

Slide 3 is ready for you to build your presentation.

TO MAKE BULLETS

Select indent before pasting or writing



This example shows how an image can span over two columns using this caption to make it a section all the way across

COLUMNS

There are three columns in this template to use and you can move around the red bar for each section with its title and text box.

FOCUSING BETATRON RADIATION ensures all photons emitted by the source passes through a homogeneous region of warm dense samples for accurate xanes measurements.

M. Berboucha (1, 2) – meriameb@slac.Stanford.edu , C. Colgan (2), E. F. Cunningham (1), P. Heimann (1), B. Kettle (2), P. King (3), HJ Lee (1), N. Lemos (3), E. Los (2), B. Nagler (1), F. Albert (3), G. Dyer (1), E.C. Galtier (1), S.P.D Mangles (2)

1. SLAC National Accelerator Laboratory, USA. 2. If Stanford faculty/staff is mentioned, Stanford University. 3. Additional organizations with logos below.

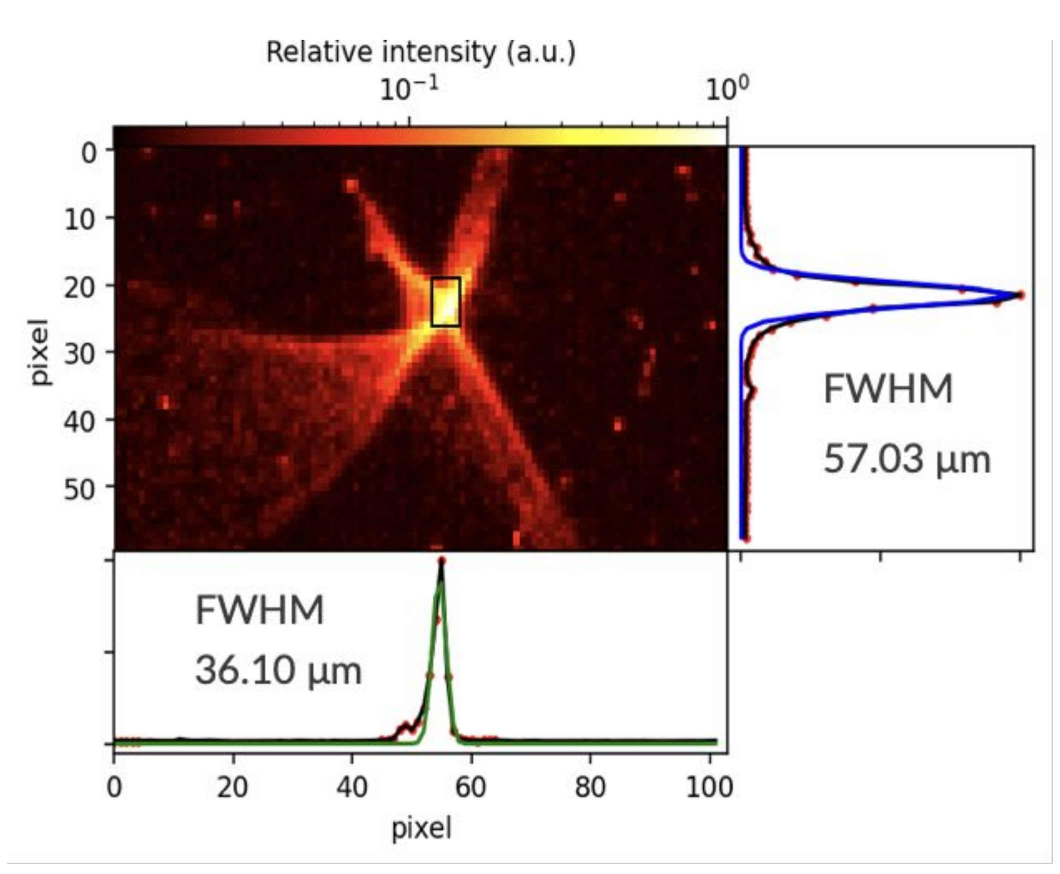
BETATRON FOR WARM DENSE MATTER STUDIES

Betatron radiation is divergent as a function of photon energy

When used as a probe for x-ray absorption spectroscopy like XANES for warm dense matter (WDM) studies there are a few problems to solve:

- Avoid probing cold material
- Increase flux in heated region
- Aperture the betatron sources that photons of energy of interest probe the sample

BETATRON FOCUSING WITH TOROIDAL MIRROR

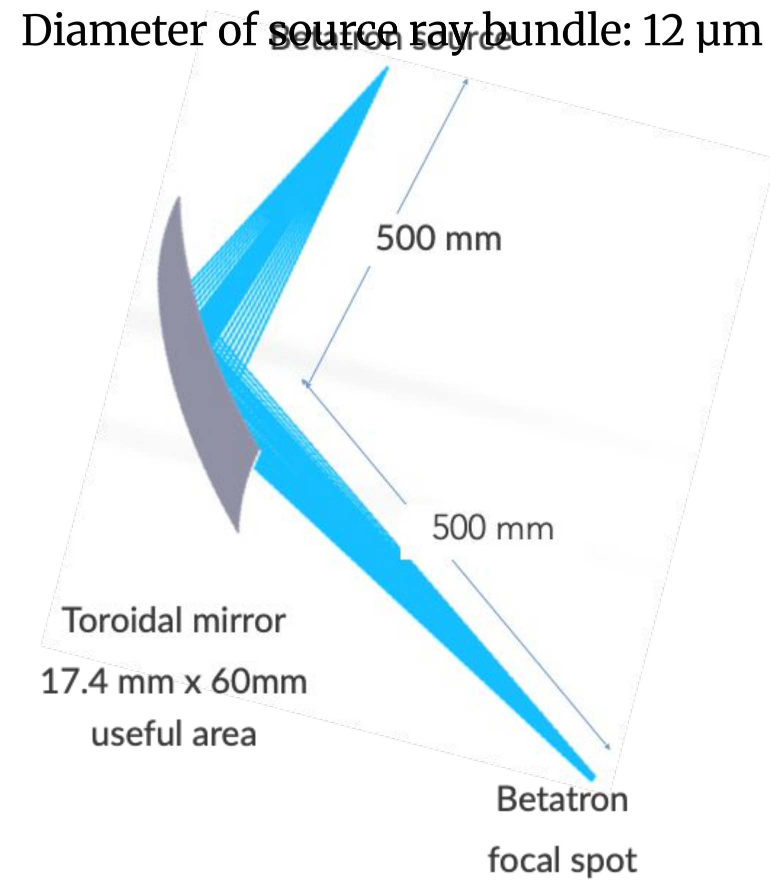


Moving along focal position

Experimental image of astigmatism through focus of betatron spot.
Observe unwanted wings In focal spot shape.
Do these wings contain high energy photons?
Aperture?

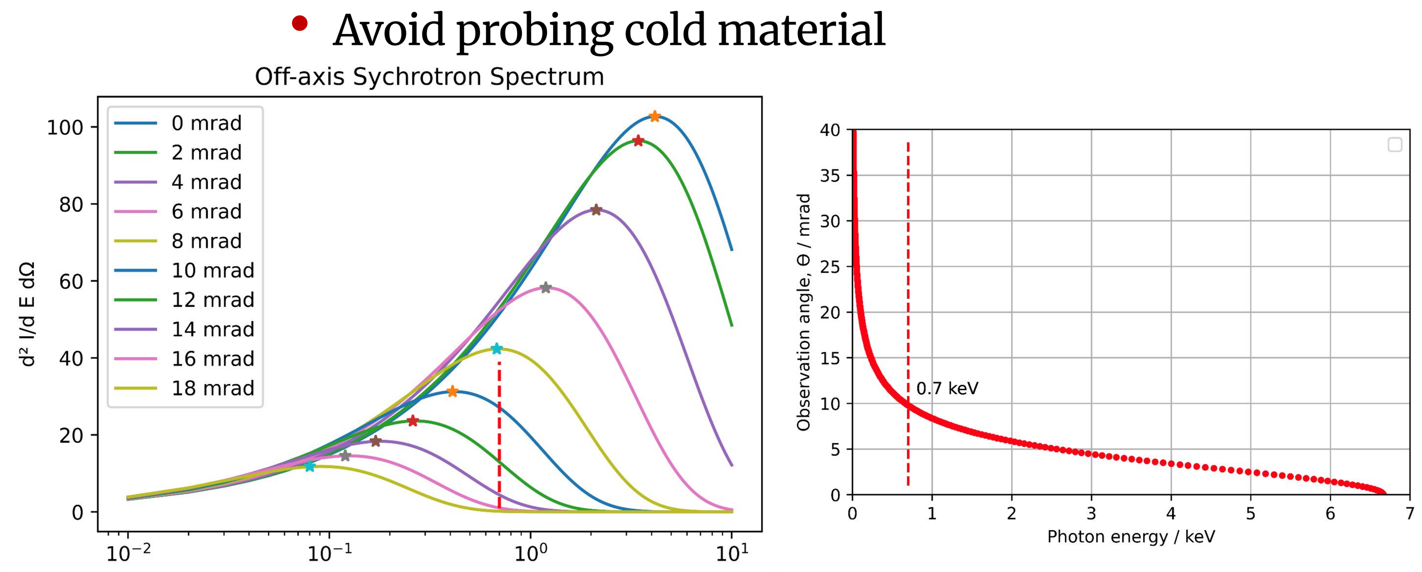
MODELLING OF BETATRON FOCUSING

- For full illumination of mirror, divergence ~ 10 mrad
- Diameter of source ray bundle: 12 µm diam

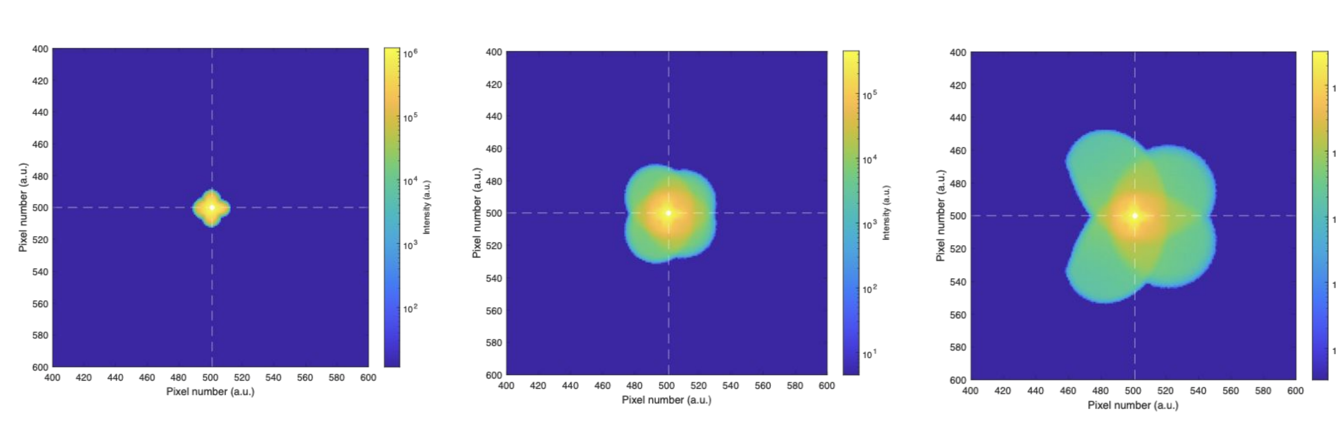


Model of betatron radiation focused with gold toroidal mirror was modelled in zemax as a function of divergence

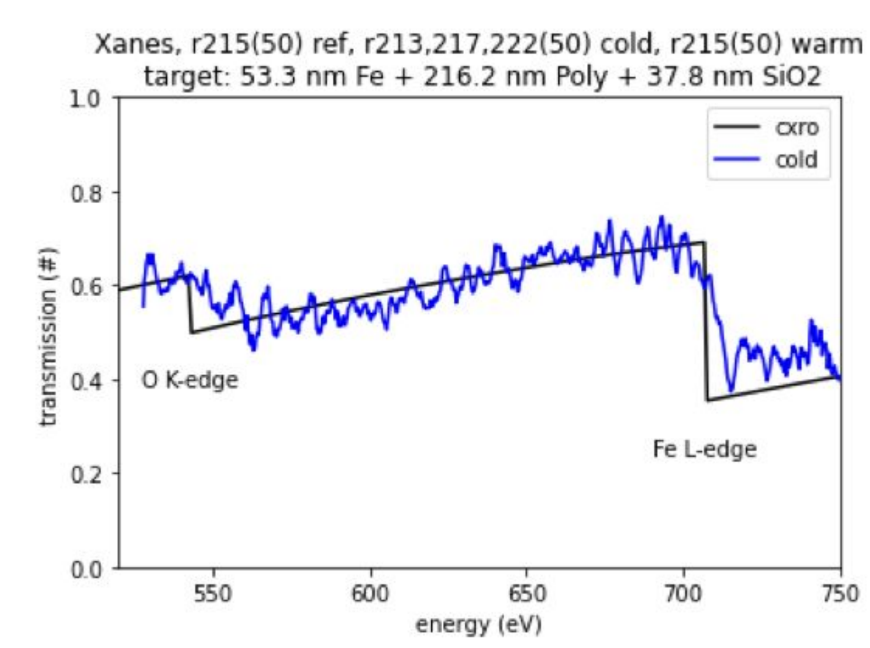
- Model beam shape as a function of divergence
- Aperture the betatron source
- Avoid probing cold material



RMS Radius: 12 µm

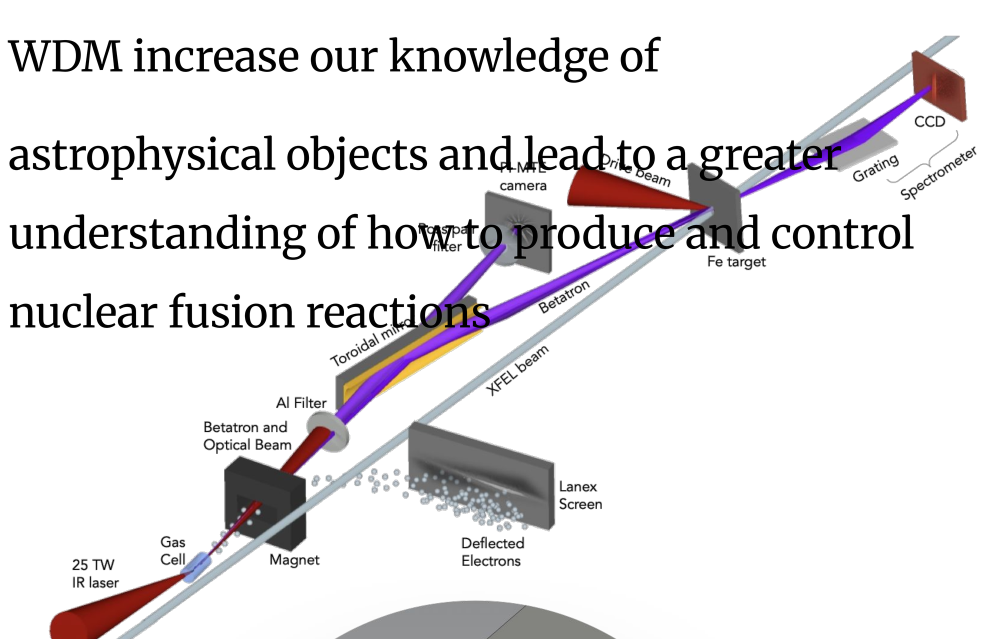


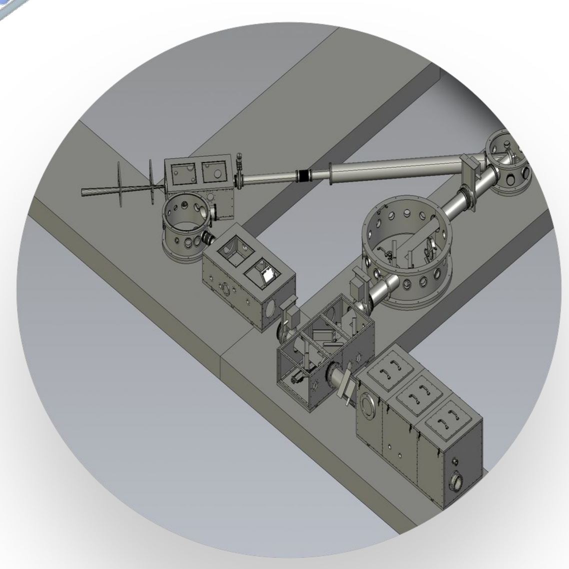
BETATRON FOCUSING FOR WARM DENSE FE STUDIES



Cold transmission of betatron spectrum through 53 nm Fe (200 nm + 38 nm SiO2) MEC LaserNetUS 2019

- Betatron focusing experiment was carried out at MEC to show capability
- Future experiment at CSU on warm dense Fe to probe the L-edge with a focused betatron source for increased SNR
- Fe is of interest since it is found in planets like Neptune and Uranus
- WDM increase our knowledge of astrophysical objects and lead to a greater understanding of how to produce and control nuclear fusion reactions





ACKNOWLEDGEMENTS

Use of the Linac Coherent Light Source (LCLS), SLAC National Accelerator Laboratory, is supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under Contract No. DE-AC02-76SF00515.

MEC end-station and itsparticipation in LaserNetUS is supported by the U.S. Department of Energy, Office of Science, Office of Fusion Energy Sciences under Contract No. DE-AC02-76SF00515.

REFERENCES

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The Process and User Interface to Streamline Cleanroom Communication and LCLS-II-HE Assembly Status

Orator Murambiwa

Mentors: Leinani Roylo, Josh Kirks

GitHub

SLAC NATIONAL ACCELERATOR LABORATORY

U.S. DEPARTMENT OF ENERGY

Stanford University

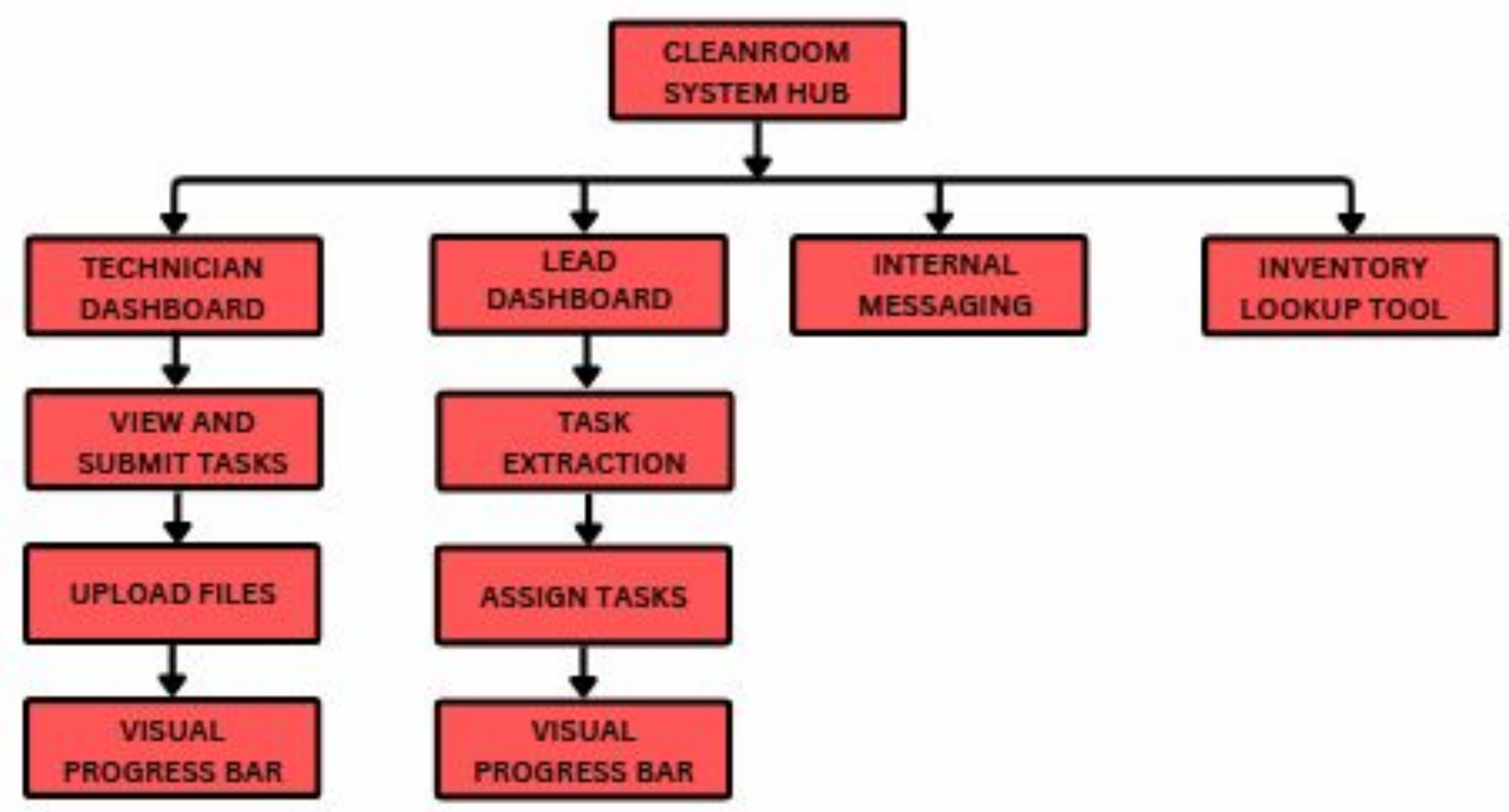
LCLS Linac Coherent Light Source

BACKGROUND

Cleanroom environments demand precise task tracking, documentation, and coordination. However, current processes rely on scattered tools, often resulting in delays, errors, and miscommunication. A unified digital system is essential to improve efficiency, accountability, and real-time visibility.

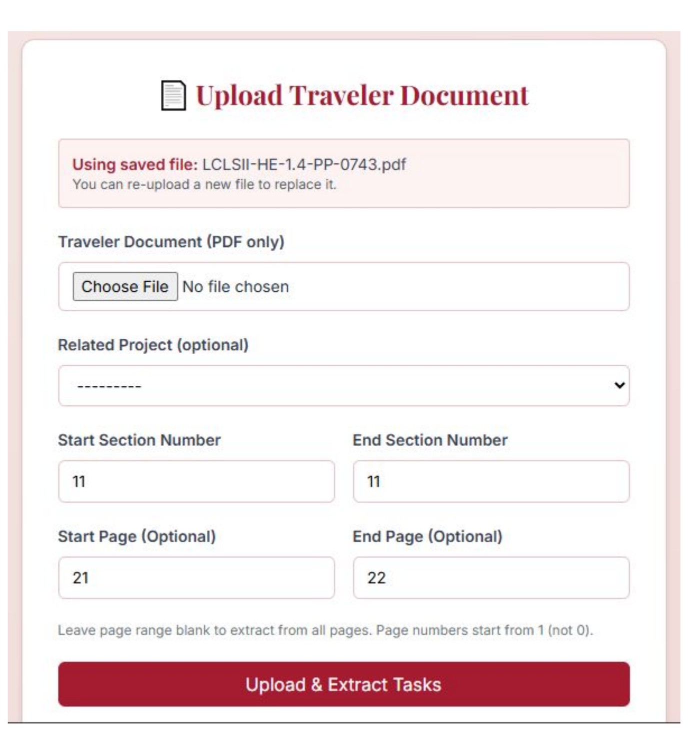
APPROACH

- Conducted user interviews with technicians and leads to guide system design
- Extracted tasks from PDF and Word docs using *pymupdf* and *python-docx*
- Built role-based dashboards, approval flows, and messaging with *django*
- Styled responsive ui components using *tailwind css*
- Parsed excel files with *pandas* for assembly parts and inventory lookup
- Added visual progress bars to track task and project completion in real time

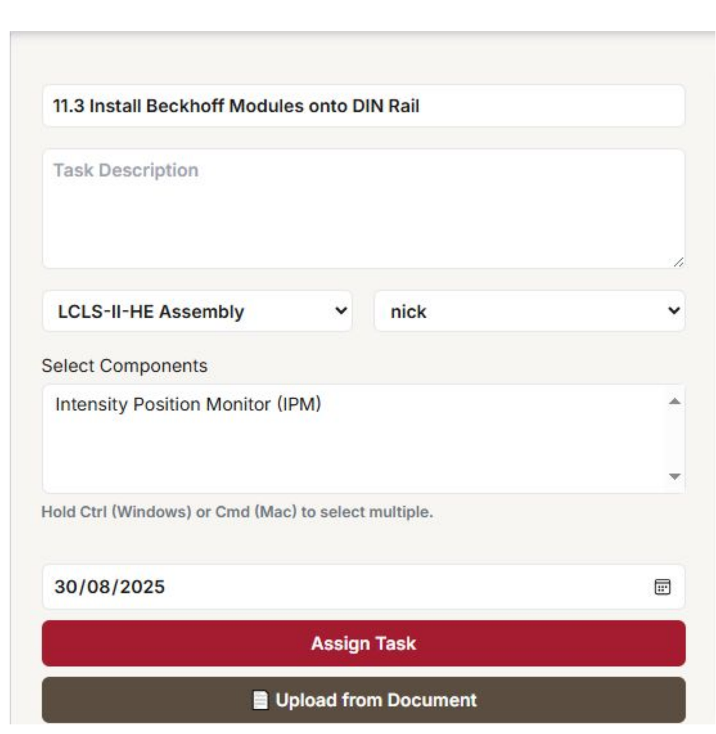


System Flowchart

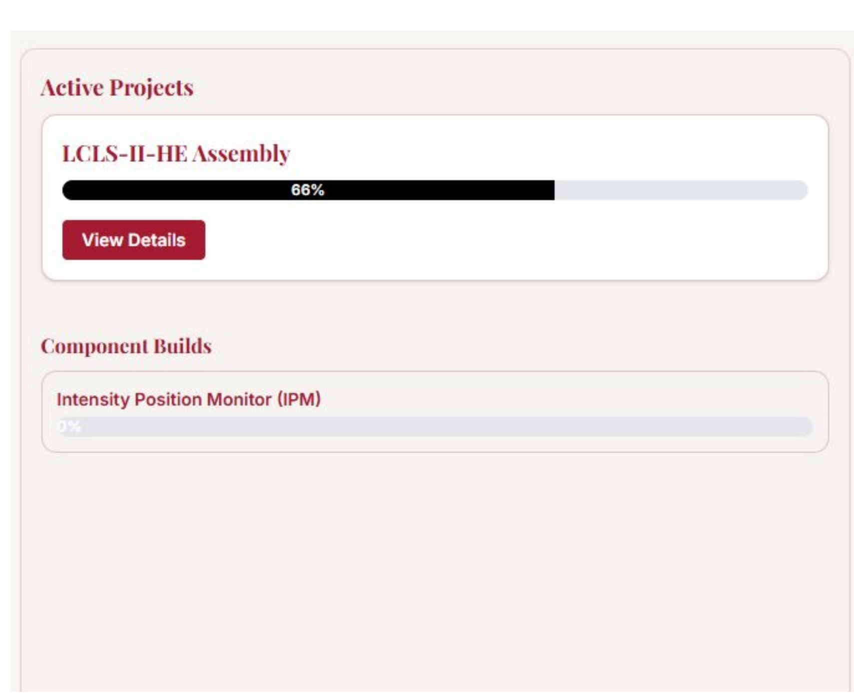
USER INTERFACE



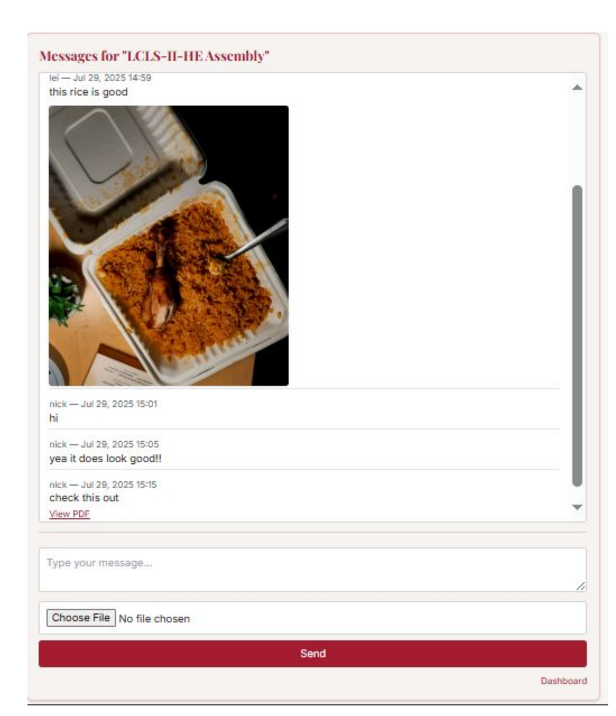
Automatic Task Extraction



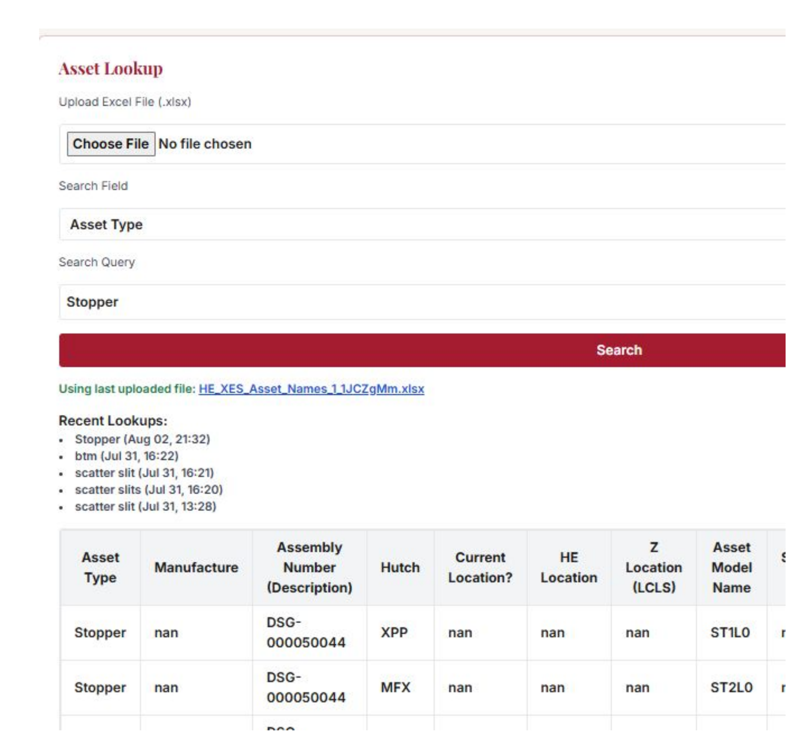
Task Assignment



Auto-Update Progress Bars



Internal Messaging Panel



Inventory Lookup Result

CODE SNIPPET

```python
for page_index, page in enumerate(pages_to_read):
    lines = page.get_text("text").splitlines()
    actual_page_number = page_range_offset + page_index + 1

    for line in lines:
        line = line.strip()

        match_section = re.match(r'^(\d+(?:\.\d+)*)\s+(.*)', line)
        match_step = re.match(r'^(\d+)\.\s+(.*)', line)
        match_sub = re.match(r'^([a-zA-Z])\.\s+(.*)', line)

        if match_section:
            sec_num, title = match_section.groups()
            try:
                if int(sec_num.split('.')[0]) in range(start_from, end_at + 1):
                    current_section = sec_num
                    current_step = None
                    headers.append({
                        'title': title.strip()[:80],
                        'description': f"{sec_num} {title}",
                        'section': sec_num,
                        'page': actual_page_number
                    })
            except:
                continue
```

Task Extraction Logic

Category	Before (Manual)	After (System)
Task assignment	Manual and time-consuming	Automated from documents
Task tracking	Spreadsheet-based	Real-time dashboard with progress bars
Inventory lookup	Manual Excel Search	Integrated search tool
Communication	Verbal or email	In-app messaging

FUTURE WORK

- Integrate an AI assistant to answer user queries from traveler documents and inventory files
- Generate automated reports (PDF/CSV) for project leads or lab audits
- Explore gesture navigation (touchless) for environments where gloves limit touch screens

ACKNOWLEDGEMENTS

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REFERENCES

[1] LCLS-II-HE PF2L2 Wavefront Sensor Assembly Traveler. Document No. LCLSII-HE-1.4-PP-0743-R1. SLAC National Accelerator Laboratory. Rev. R1, Released December 17, 2024.

[2] Dauzon, Samuel, Aidas Bendoraitis, and Arun Ravindran. *Django: web development with Python*. Packt Publishing Ltd, 2016.